

DocuChat AI + Quiz

Product Specification & Project Roadmap

1. Product Overview

Type: Web-based AI learning assistant

Goal: Enable users to interact with documents through a dual-interface system featuring a Chat Mode for exploration and a Quiz Mode for knowledge verification.

2. Problem Statement

- Reading and understanding large PDFs is time-consuming.
- Users struggle to retain key information after passive reading.
- Traditional tools lack interactive learning components.
- Standard LLMs may hallucinate without document-specific grounding.

3. The Solution

A RAG-based (Retrieval-Augmented Generation) system that retrieves relevant document content to generate accurate answers and creates customized quizzes to reinforce learning.

4. Target Audience

- **Students:** For exam preparation and active recall.
- **Researchers:** For rapid analysis of academic papers.
- **Professionals:** For efficient learning from corporate reports.

5. Core Features

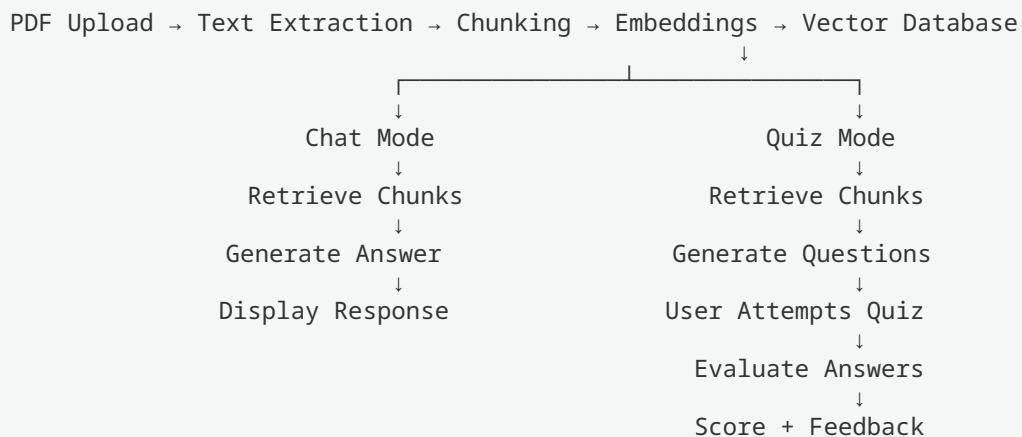
Chat Mode

- ✓ Context-aware Q&A based on uploaded PDFs.
- ✓ Precise source citation (including page numbers and text snippets).

Quiz Mode

- ✓ Automatic quiz generation from document context.
- ✓ Multiple Question Types: MCQs, True/False, and Short Answer.
- ✓ Performance evaluation with scores and detailed feedback.

6. Technical Architecture & Flow



7. Functional Requirements

- **PDF Processing:** Multi-page text extraction.
- **Retrieval System:** Semantic similarity search using top-K relevant chunks.
- **Feedback System:** AI-generated explanations for correct answers and improvement tips.

8. Technical Stack

- **Frontend:** Streamlit / React
- **Backend:** Python / FastAPI
- **Orchestration:** LangChain
- **Vector DB:** FAISS
- **LLM:** OpenAI GPT series

9. Project Timeline (5 Weeks)

Week	Focus Area
Week 1	PDF Parsing & Text Chunking Logic
Week 2	Embedding Generation & Vector DB Integration
Week 3	Chat Mode & RAG Pipeline Implementation
Week 4	Quiz Generation Engine & Evaluation Logic
Week 5	UI Refinement, Testing & Deployment

10. Resume-Ready Description

"Developed a RAG-based PDF chatbot with an integrated AI-powered quiz system that generates context-aware questions and evaluates user responses. Implemented semantic retrieval, source-grounded answers, and interactive learning features using LangChain and FAISS."